

Jared J. Schultz

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EDUCATION

University of Minnesota, College of Science and Engineering *Minneapolis, MN*
Master's of Science, Robotics, Minor in Mechanical Engineering 9/2024 – 5/2026
GPA: 4.00
University of Minnesota, College of Science and Engineering *Minneapolis, MN*
Bachelor of Science, Computer Engineering 9/2020 – 5/2024
GPA: 3.52

TECHNICAL SKILLS

Robotics: Computer Vision, Machine Learning, AI, Path Planning, Kalman Filtering, Feedback Control, Force Feedback
Programming: C, C++, Java, Python, ROS, Bash, MATLAB, Verilog
Mechanical: SolidWorks, RoboDK, 3d Printing (SLA and FDM), Machine shop trained 100+ hours (mill, lathe, waterjet)
Electrical: DC/AC circuit analysis, Altium PCB Designer, Vivado FPGA
Tools and Platforms: Git, Jira, Confluence, Linux, NumPy, PyTorch, TensorFlow

JOB AND RESEARCH EXPERIENCE

University of Minnesota and Medtronic Inc. *Minneapolis, MN*
Mechatronic Lead for Autonomous Telerobotic Surgical Stroke Robot 6/2025 – 5/2026

- Collaborated with neurosurgeons, engineering faculty, and Medtronic leadership to define system requirements for a cost-efficient robot able to perform remote thrombectomy.
- Integrated mechanical, embedded, and AI-driven subsystems into a unified robotic architecture for remote guidewire navigation. Live Demo: telebot.cse.umn.edu
- Built an automated test platform using an ATI Nano17 6-DOF force/torque sensor and a stepper-actuated test rig to characterize prototype performance, accelerate wear, and identify failure modes to promote device longevity.
- Designed, manufactured, and validated multiple robot prototypes using custom SLA printed gears, hypoid transmissions, and differential mechanisms, using experimental results to drive design improvements.
- Designed and implemented a motor control architecture that decouples degrees of freedom and enables precise speed and position control of the 3-DOF rolling contact mechanism.

General Dynamics Mission Systems *Bloomington, MN*
Software Engineering Intern 5/2022 – 8/2022

- Translated the existing test environment from Bash into Python to improve readability and package support.
- Traced bugs through layers of code and wrote tickets for unresolved issues with detailed reproduction steps.
- Anticipated future requirements and implemented features into the test environment to promote code longevity.
- Determined software requirements and developed system tests to validate the state of the project.
- Contributed to professional documentation through Confluence, understood team workflow through Jira, and streamlined version control through GitLab.

PUBLICATIONS

Friction Characterization of a Roller-Drive Mechanism for Robotic Motion Control of Guidewires DMD 2026
Jared Schultz, Pin-Hao Cheng, Matt Rajala, Timothy M. Kowalewski [PDF] doi.org/10.1115/DMD2026-1086

Towards Remote Thrombectomy with Telerobotically-Driven Guidewires DMD 2026
Pin-Hao Cheng, Ronak Narkhede*, Matt Rajala*, Jared Schultz*, Nathan Harbinson, Samuel Fisher, Nitish Poojari, Sharva Khandagale, Alex Berg, Keara Berlin, Adam Imdieke, Michael Feldkamp, Scott Frushour, Kaustubh Patil, Mark Ashby, William Peine, Sean L. Moen, Andrew Grande, Karthik Desingh, Timothy M. Kowalewski.* [PDE] doi.org/10.1115/DMD2026-1079

* Contributed equally, listed alphabetically by last name

PROJECT EXPERIENCE

- Robotics Club projects (ARC Robotics, NASA Lunabotics, Robot in Three Days)** 1/2021 – Present
- Work under a three-day time crunch to design and manufacture a robot to meet competition requirements.
 - Use machine learning and vision tracking to improve the accuracy of the manipulator and perform localization.
 - Create a series of professional videos and a white paper documenting our engineering processes.
- Computer Vision and ML system for the ARC Robotics Project (Class Number: CSCI 5561)** 1/2025 – 6/2025
- Created a custom neural stack that identifies, tracks, and predicts the movement of robots in a moving-camera, moving-target environment.
 - Achieved 98% classification accuracy with a 3ms loop time, successfully fulfilling the design requirements.
 - Optimized the developed solution to run in real time on an Nvidia Orin Nano while preserving all functionality.
- Intermediate Medical Robotics Projects (Class Number: ME 8284)** 9/2024 – 12/2024
- Implemented kinematics, self-motion, path planning, and Jacobian IK for a hyper-redundant planar robot
 - Used end-effector force measurements to implement feedback and measure the effects of delay on manipulability
 - Developed methods for control, teleoperation, and feedback for UR5 Robot over TCP/IP Ethernet
- Robot Vision Final Project (Class Number: EE 5271)** 9/2024 – 12/2024
- Designed and built from scratch an autonomous 6 DOF arm able to simulate industrial pick-and-place tasks
 - Implemented Vision algorithms to allow the robot to recognise specific objects
 - Developed and debugged a custom position-based visual servoing algorithm to allow for dynamic object tracking.
- Personal Robotics Projects** 1/2022 – 12/2024
- Design, prototype, and calculate kinematics for a 6-degree-of-freedom arm and a transforming hexapod
 - Manage memory and hardware constraints when working with the microcontrollers.
 - Manage a budget and compare hardware to determine the best actuators and sensors for the project.
- Autonomous Quadrotor for Education (Class Number: EE 4951)** 12/2023 – 5/2024
- Assisted in exploratory research, completing the initial steps in creating a new class for autonomous aerial flight.
 - Communicated with software vendors to troubleshoot and resolve system bugs.
 - Implemented signal processing and path planning algorithms in a pipeline, allowing the real-time position of the drone to be used to follow a path autonomously.
- Intelligent Robotic Design Final Project (Class Number: CSCI 5551)** 1/2023 – 5/2023
- Designed and implemented a Multi-Robot Search using SLAM and RRT to be run in a classroom setting.
 - Bug-tested networking protocol issues and physical limitations in the provided robotic system

RELEVANT CLASS EXPERIENCE

Completed classes at the University of Minnesota, Twin Cities

ME 8284 Intermediate Robotics with Medical Applications	CSCI 8551 Intelligent Agents
ME 5286 Robotics	CSCI 5801 Software Engineering 1
ME 5281 Feedback Control Systems	CSCI 5607 Computer Graphics 1
BMEN 5601 Cardiovascular Devices	CSCI 5551 Introduction to Intelligent Robotic Design
BMEN 5413 Neural Interfacing	CSCI 5561 Computer Vision
BMEN 5151 Intro BioMEMS/Medical Microdevices	CSCI 5521 Machine Learning Fundamentals
EE 5271 Robot Vision	CSCI 5512 Artificial Intelligence 2
EE 5321 Linear Control Systems	CSCI 5511 Artificial Intelligence 1

LEADERSHIP EXPERIENCE

<i>Robotics Club</i>	<i>Minneapolis, MN</i>
Robomaster Project Founding Member, Mechanical Captain	6/2024 – Present
<i>Triangle Fraternity MN Chapter</i>	<i>Minneapolis, MN</i>
Vice President of Internal Affairs, Project Manager, Ritual Keeper	1/2022 – 12/2026
<i>First Robotics</i>	<i>Buffalo, MN</i>
First Robotics Lead Programmer, Team Captain	1/2018 – 3/2020
<i>Boy Scouts</i>	<i>Buffalo, MN</i>
Eagle Scout, Senior Patrol Leader, Troop Guide	4/2015 – 8/2018